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Prefix Sum

	0	1	2	3	4	5	6	7
arr	7	1	2	4	3	2	5	4

	0	1	2	3	4	5	6	7
pre	7	8	10	14	17	19	24	28

	(l, r)
17	(0, 4)
20	(2, 7)
9	(2, 4)
14	(3, 6)
⋮	⋮

m - queries

Statement: n-sized array is given & 'm' queries are given

M-1 : Brute Force - loop from l to r T.C. = $O(m * n)$

M-2 : Prefix Sum
↓
running sum

T.C. = $O(m + n)$

pre[i] = pre[i-1] + arr[i]
↓
(0, i) tak ka sum

A.S. = $O(n)$

(2,7) = (0,7) − (0,1)

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Q// Equilibrium Point

[-7, 1, 5, 2, -4, 3, 0]

pre = -7 -6 -1 1 -3 0 0

suf = 0 7 6 1 -1 3 0

Ex: arr = 1 2 3 4 5 10 (i = 4, j = 10)

sumi = ~~1~~ ~~3~~ ~~6~~ 10

sumj = 10

Why is this method wrong

Ml: Brute Force TC = $O(n^2)$

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Sieve of Eratosthenes

~~1~~	2	3	~~4~~	5	~~6~~	7
~~8~~	~~9~~	~~10~~	11	~~12~~	13	~~14~~
~~15~~	~~16~~	17	~~18~~	19	~~20~~	~~21~~
~~22~~	23	~~24~~	~~25~~	~~26~~	~~27~~	~~28~~
29	~~30~~	31	~~32~~	~~33~~	~~34~~	~~35~~
~~36~~	37	~~38~~	~~39~~	~~40~~	41	~~42~~
43	~~44~~	~~45~~	~~46~~	47	~~48~~	~~49~~

upto 49, no. of prime numbers

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Q// Count primes in range

(3,10)

sieve =

0	1	2	3	4	5	6	7	8	9	10
0	0	1	1	0	1	0	1	0	0	0

count = 1+1+1 = 3

pre =

0	1	2	3	4	5	6	7	8	9	10
0	0	1	2	2	3	3	4	4	4	4

0 - composite

1 - prime

** if multiple queries are there, use PREFIX SUM.